Brainstorm & Idea Prioritisation

Date	12-06-2026
Team ID	
Project Name	**Voice Based Concept Understanding Analyser**
Maximum Marks	4 Marks

Brainstorm & Idea Prioritisation Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem-solving. Prioritising volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions

Step-1: Team Gathering, Collaboration and Selecting the Problem Statement

As this project was developed individually, I conducted a structured brainstorming session by identifying common challenges faced by students during examinations, viva voce, and technical interviews. I analysed existing AI learning tools and observed that most platforms evaluate written responses but do not assess a student's spoken conceptual understanding. **After reviewing existing solutions and identifying gaps, I finalised the following problem statement:**

Students often struggle to determine whether they truly understand a concept when explaining it verbally. Existing educational platforms primarily evaluate written responses and provide limited feedback on conceptual understanding, communication skills, and spoken explanations. There is a need for an AI-powered system that can automatically analyse spoken explanations, assess conceptual accuracy, and provide personalised feedback to improve learning outcomes.

Why was this problem selected?

- Addresses a real educational challenge.
- Useful for students preparing for interviews and viva examinations.
- Combines Speech AI and Generative AI.
- Provides measurable educational impact.
- Feasible using modern AI technologies (Whisper, Sentence-BERT, Gemini).

Step-2: Brainstorm, Idea Listing and Grouping

AI Evaluation

- Semantic similarity
- Keyword matching
- AI tutor feedback Gemini integration

Speech Analysis

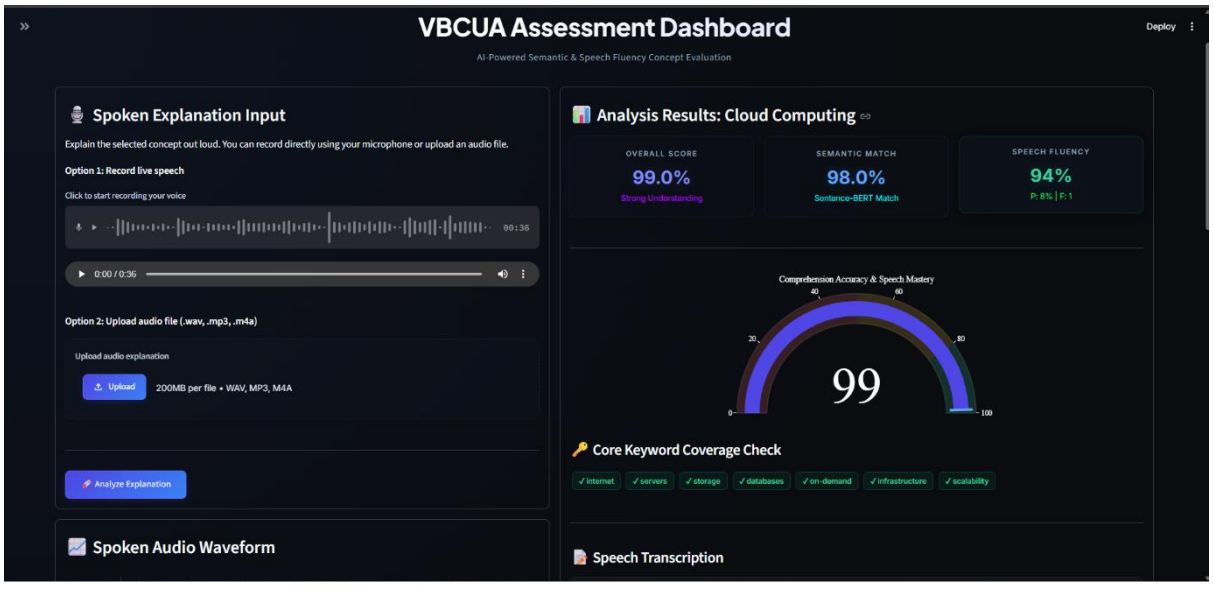
- Fluency score
- Pause detection
- Filler words
- Speaking volume

Reporting

- Dashboard
- Historical reports
- PDF generation
- Progress analytics

User Experience

- Login
- Concept selection
- Audio recording
- Result visualisation



Step-3: Idea Prioritisation

High Importance + High Feasibility

- Speech-to-text transcription
- Semantic similarity
- AI-generated feedback
- PDF reports
- Dashboard
- Keyword analysis

High Importance + Medium Feasibility

- Pronunciation analysis
- Confidence estimation
- Faculty analytics

Medium Importance + High Feasibility

- Gamification
- Dark mode
- Leaderboards

Low Priority

- Mobile app
- Social sharing

